	BELLOWS CONTROLS SERVIÇOS AERONÁUTICOS.COM 201712-31/ANACAssembly Fuel Control Unit		
Fuel Control Unit	P/N: 4138008 / 3244872	S/N:	
Model: DP-F2	Data:	O/S: BEL	
Client:	Pag: 1 de 20		
Manual*: 3244872	Rev. 2	Date: Nov. 18, 2020	ATA: 73-20-64

	Assembly of Drive Body Assembly			
01	Assemble the drive body assembly as follows (Ref. IPL Fig. 1):CAUTION: A HIGH DEGREE OF CLEANLINESS WHILE HANDLING ANDINSTALLING BALL HEARINGS IS VITAL. CARE SHOULD BE TAKEN TO AVOID ANY PRESSURE ON SHIELDED SURFACES, AS THESE SURFACES ARE VERY SOFT AND DAMAGE WILL LIMIT BALL BEARING LIFE.CAUTION: BALL BEARINGS (370 AND 385) ARE OPEN ONE SIDE AND SHIELDED THE OTHER, AND MUST BE INSTALLED WITH THE SHIELD FACING AWAY FROM SPACERS (375 AND 380) (REF. FIG. 701).	702	CUIDADOS	CUIDADOS

02	A. Assemble new ball bearing (385, IPL Fig.1) into drive body (390) with shield side facing away from spacers as shown in Fig. 701. Install spacers (375 and 380, IPL Fig.1) and fill the annuls with 0.2 to 0.3 grams (0.007 to 0.011 oz.) of grease (2, Table 704). Assemble new ball bearing (370, IPLFig.1) with the shield side facing away from spacers as shown in Fig. 701. Assemble plate (360, IPL Fig.1) and secure with screws (365). Torque screws to value specified(Ref. Table 804).	702		
03	B. Install new ball bearing (235, IPL Fig.1) into speed bias lever (245) and secure with new pin (240). Assemble new ball bearings (210) into governor lever assembly (215). Assemble speed bias lever assembly (230) into governor lever assembly(215) and install new pin (205).	702		

04	C. Install speed bias spring (180), adjustment screw (175), washer (170) and new nut (165).NOTE: Do not tension speed bias spring (180) at this point of assembly.CAUTION: IF EXCESSIVE PRESSURE IS APPLIED BY PIVOT SCREW (195), GOVERNOR LEVER ASSEMBLY (215) WILL SUSTAIN PERMANENT DAMAGE.	702		
05	D. Position lever assemblies into drive body (390) and assemble washer (190) and pivot screw (185). Torque pivot screw (185) to value specified (Ref. Table 804). Assemble shim(s) (200) and pivot screw (195). Do not torque pivot screw (195) at this stage of assembly. Mount drive body on drive body holding fixture (13, Fig.901).	705		

06	E. Check governor lever assembly (215, IPL Fig.1) for side play as follows (Ref. Fig. 702):(1) Install dial indicator and bracket (I) to fixture (13, Fig. 901). Position needle on dial indicator (1, Fig.702) against governor lever assembly(5).(2) Move governor lever assembly (5) and observe movement on dial indicator (1).(3) Add or subtract shim(s) (3) until a movement of 0.003" to0.006" (0.008 cm to 0.015 cm) is obtained.	705		
07	(4) Torque pivot screw:(4) progressively, at the same time checking side playoff governor lever assembly (5). Repeat steps (2) and (3), preceding, until correct clearance is obtained with pivot screw (4) torqued (Ref. Table 804).(5) When correct clearance is obtained, removed dial indicator and bracket (1).	705		

08	F. Lubricate new preformed packings (130 and 140, IPL Fig.1) with oil (1, Table 704) and assemble to air flow restrictors (125 and 135, IPL Fig.1). Assemble air flow restrictors (125 and 135) into drive body (390) until flush with inside face of drive body.NOTE: An adhesive patch is bonded to (Py) air flow restrictor (135) threads. Replace (Py) air flow restrictor (135) if it turns with a torque less than the value specified (Ref. Table 804).	705		
09	G. Preset and test (Py) air flow restrictor (135) as follows (Ref. Fig. 703):(1) Attach orifice preset adapter (5) to drive body (7) and connect bleed adapter fixture (4) as illustrated.	705		
10	(2) Install orifice adjusting fixture (6) on adapter mounting face of drive body (7).(3) Apply drive body preset weight (9) to governor lever assembly (11).NOTE: Load must hang at 90 degrees to lever.	705		

11	(4) Apply air pressure at 75 ± 2 psig (517 ± 14 kPag) and adjust (Py)air flow restrictor (135, IPL Fig.1) until pressure drop across bleed adapter (4, Fig. 703) is between zero and 10.0 inches (25.4 cm) of Kerosene.NOTE: Leakage limit shall be established by adjusting orifice clockwise from excessive leakage position (fully out) until lowest pressure differential across bleed adapter is obtained.	708		

12	(5) Remove drive body preset weight (9) and apply drive body preset weight (10) to governor lever assembly (11).Pressure differential across bleed adapter fixture (4) shall be within 5 inches (12'.7 cm) of Kerosene of reading obtained with drive body preset weight (9). Reset, if necessary, to obtain differential within 5 inches (12.7cm) of Kerosene.	708		

13	(6) Remove orifice adjusting fixture (6). Install new gasket (105, IPL Fig.1). Align (Py) air flow restrictor (135) to nearest serration of gasket (105), if necessary.Recheck leakage as specified in step (4), preceding, if air flow restrictor (135) is realigned.(7) Remove gasket (105) and retain for future use.special tools, fixtures and equipment.	708		
14	H. Preset and test (Px) air flow restrictor (125) as follows (Ref. Fig. 704):(1) Install fluted pin (5) into (Px) air flow restrictor (4).Seat valve (1) on fluted pin (5).(2) Apply drive body preset weight (8) to governor lever assembly (9).(3) Position speed bias lever assembly (7)so that gap between governor lever assembly (9) and speed bias fewer assembly (7) is to top.	708		
15	(4) Adjust air flow restrictor (4) inwards until valve (1) is not seated on (Px) air flow restrictor. Install dial indicator and bracket (2), valve setting fixture (3), and position dial indicator needle on top of valve setting fixture (3). Set dial indicator to zero.	708		

16	(5) Adjust (Px) air flow restrictor (4) outwards until 0.003" to 0.004" (0.008 cm to 0.010 cm) travel is indicated on dial indicator (2).NOTE: Gap between levers shall be maintained at the top during adjustment.	708		
17	(6) Position speed bias lever assembly (7) so that gap between lever assemblies is to bottom. Install gasket (105, IPL Fig.1). Observe travel of valve (1, Fig.504), as indicated on dial indicator (2). Total travelindicated shall be between 0.0075" to 0.0105" (0.0191 cm to 0.0267 cm) and must be maintained after (Px) air flow restrictor (4) is aligned with nearest serration of gasket (105, IPL Fig.1).(7) Remove special tools, fixtures and equipment.	710		

18	I. Install air flow jets (95 and 100) to adapter (80) using special screwdriver (46, Fig. 901) and torque to value specified (Ref. Table 804). Install spring (110, IPL Fig.1) onto valve (115) and assemble adapter (80). Secure with washers (90) and screws (85). Torque screws to value specified (Ref. Table 804).J. Adjust new nut (165, IPL Fig.1) until three threads protrude through nut.	710		
19	K. Perform drive body preflow test as follows (Ref. Fig. 705):(1) Plug threaded outlets in orifice preset adapter (10) and install onto drive body (6). Plug Py pressure port (14) in adapter (4).(2) Connect bleed adapter (3) across Pc-Py differential manometer (12) in line from air pressure source (1).	710		
20	(3) Complete test rig installation as illustrated in Fig. 705 and connect to Px pressure port (13) in adapter (4}.(4) Attach drive body preset weight (7) to governor lever assembly (8) and introduce air pressure at 40 psia (276 kPa) to bleed adapter (3). Differential pressure across bleed adapter (3) must be between 10.8 and 12.0 inches (27.4 to 30.5 cm) of mercury.	710		

21	(5) If differential pressure is not within limits, select (Pa) air flow jet (11) (sizes 64D to 67D) to obtain required differential pressure.(6) Close speed bias valve by transferring gap (View 'A, 15, Fig. 705) between levers (8 and 9) to the top surface (View 'B', 15, Fig. 705), using hand pressure on speed bias lever assembly (9). The differential pressure across the speed bias valve must not exceed 2.8 inches (7.1 cm) of mercury.(7) Remove all special tools, fixtures and equipment.	710		

22	L. Remove governor and speed bias lever assemblies (215 and 230, IPL Fig.1) from drive body (390).M. Assemble new tube (325) to drive shaft (320) and check dimensionally as specified in CHECK, paragraph 2.B. Assemble governor weights (355) and washers (350) into drive shaft assembly (315) and install new pins (345). Secure the pins with clip (340). Assemble washer (330) onto drive shaft assembly (315). Insert drive shaft assembly (315) into drive body (390) and secure with washer (310) and a slave nut equivalent in size to nut (305). Using weight table holding fixture (15, Fig. 901) to immobilize the drive shaft, torque the slave nut to value specified for nut (305, IPL Fig.1) (Ref. Table 804).NOTE: The outside surface of tube (325, IPL Fig.1) must be free from tool marks after assembly.	712		
23	N. Assemble bearing assembly (285, IPL Fig.1) as follows:(1) Install new ball bearing (295) into bearing cap (290).(2) Assemble ball bearing and bearing cap onto bushing assembly (300).	712		

24	O. Shim drive shaft assembly (315) as follows (Ref. Fig. 706):(1) Install bearing assembly (6) onto drive shaft assembly(1).	712		
25	NOTE: If the area of drive shaft assembly (1), where the lug of bushing assembly (300, IPL Fig.1) a buts, is dented, ensure that the lug is positioned on the undented side of drive shaft assembly (1, Fig. 706).	712	ADVERTÊNCIA	ADVERTÊNCIA
26	(2) Reinstall the governor and speed bias lever assemblies as described in paragraphs 2.C. and D., preceding.(3) Suspend drive body preset weight (7) from governor lever assembly (9) and check clearance between bearing assembly (6) and ball bearing (8).NOTE: Ensure bearing assembly (6) is bottomed on drive shaft assembly (1) and face of speed bias lever assembly (12) contacts face of governor lever assembly (9) when measuring clearance.	712		

27	(4) Calculate thickness of shim(s) (2) required [recorded clearance less 0.003" to 0.007" (0.008 cm to 0.018 cm)].(5) Remove governor and speed bias lever assemblies (9 and12}.(6) Remove slave nut (11) and drive shaft assembly (1).(7) Add calculated amount of shims [2) between washer (3) and clip (5).	712 / 714		
28	(8) Reinstall drive shaft assembly (1) into drive body (4) asper paragraph 2.M., preceding.(9} Reinstall lever assemblies (9 and 12) and recheck clearance as per step (3).(10) Reshim as per steps (1) through (9), preceding, if required.(11) Measure drive shaft assembly (1) length for conformance with dimension indicated in Fig. 706. If spline leng this not within limits, adjust by resetting (Py) air flow restrictor (135, IPL Fig.1) and reshimming drive shaft assembly (1, Fig. 706). Replace drive shaft assembly if limits cannot be attained.	714		

29	P. Remove slave nut from drive shaft assembly (315, IPL Fig.1) and install new nut (305). Use weight table holding fixture (15, Fig. 901) to immobilize drive shaft assembly and torque nut to value specified (Ref. Table 804).	714		
30	NOTE: Lubricate OD of eccentric shaft (275, IPL Fig.1) and ID of governor cam lever (280) with grease (2, Table704) prior to assembly.	714	AVISO	AVISO
31	Q. Install governor cam lever (280) into drive body (390) and assemble eccentric shaft (275). Rotate eccentric shaft (275) into drive body (390) until the head is flush with the drive body, and the scribe line on eccentric shaft (275) coincides with the raised mark on drive body (390). Assemble washer (270) and nut (265). Torque nut to value specified (Ref. Table 804).	714		

32	R. Attach governor spring (160A, IPL Fig.1) to governor cam lever (280). Attach adjustment screw (155) to the other end of governor spring (160A) and connect to governor lever assembly (215). Install washer (1501 and new nut (145) until one complete thread of adjustment screw (155) protrudes through nut (145).S. Install adjustment screw (260), washer (255) and nut (25.0) to drive body (390).	714		
33	T. Adjust governor spring (160) tension as follows (Ref. Fig.707):(1) Mount drive body (5) on governor spring preset plate (7)NOTE: Install governor spring preset plate (7) onto governor spring preset fixture (14, Fig. 901).	716		

34	(2) Connect an ohmmeter between governor spring preset plate insulated pin (6, Fig. 707) and governor cam lever (1).(3) Connect governor cam lever (1) to governor spring preset fixture (14, Fig. 901) spring tensioner.(4) Lightly tension governor spring (2, Fig. 707) by tightening nut (4), ensuring that the ohmmeter is reading zero [governor cam lever (1) and insulated pin (6) are making contact].(5) Turn governor spring preset fixture (14, Fig. 90,1) spring tensioner adjustment nut until the ohmmeter reads infinity (contacts just open). Check indicated spring tension.	716		

35	(6) Adjust governor spring tension as in steps (4) and (5), preceding until a tension of 4.4 pounds (211 Pa) is established with contacts just opening. Tap governor spring (2, Fig. 707) to ensure no false tension is applied.NOTE: If unable to achieve the required spring tension, remove nut (4), add another new washer (3) (maximum of two washers), reassemble nut (4) and repeat steps (2) through (6), preceding.(7) Remove drive body (5) from governor spring preset plate (7).	716		
36	U. Place drive body assembly (60,, IPL Fig.1) into a clean container, free from dust, grit and moisture.	716		
37	Assemble flow body assembly as follows (Ref. IPL Fig. 1):NOTE: Use PowRarm (2, Fig. 901), PowRarm adapter (40) and mounting bolt (39) to facilitate assembly.A. Lubricate new preformed packings (1450, 1460 and 1470, IPL Fig.1) with oil (1, Table 704) and assemble onto plugs (1445, 1455 and 1465, IPL Fig.1). Screw plugs into main body assembly (1475) and torque to value specified (Ref. Table 805}.	716		

38	B. Assemble and install bypass valve assembly (1385, IPL Fig.1) as follows:(1) Lubricate new preformed packings (1430 and 1435) with oil (1, Table 704) and assemble onto bypass valve sleeve assembly (1425, IPL Fig.1). Assemble bypass valve sleeve assembly (1425) into main body (1505).(2) Assemble new diaphragm (1405), diaphragm retainer (1400), washer (1395) and nut (1390) onto bypass valve (1410). Torque nut (1390) to value specified (Ref. Table 805).Install retainer (1415, IPL Fig.1), new gasket (1420) and bypass valve assembly (1385) into main body assembly (1475).	717		
39	(3) Assemble spacer (1380). Assemble spring (1375), spring holder (1370) and bimetallic disc group (1365) onto bypass valve assembly (1385).NOTE: Bypass valve assembly (1385) must be assembled with a standard spacer (1380) (thickness 0.054" to 0.058', (0.137 cm to 0.147 cm)).	717		

40	(4) Lubricate new preformed packing (1360) with oil (1, Table704) and assemble onto guide (1355, IPL Fig.1). Assemble guide (1355) into cover assembly (1330).Lubricate new preformed packing (1350) with oil (1, Table704) and assemble onto cover assembly (1330, IPL Fig.1).Assemble retaining plate (1320) to cover assembly (1330)and install the cover assembly into main body assembly (1475), with the scribe line on the cover towards the bellows position.	717		
41	NOTE: If leaf spring (1335) or ball (1340) were removed at DISASSEMBLY, reinstall new items into cover (1345) prior to installation of cover assembly (1330).	717	ORIENTAÇÃO	ORIENTAÇÃO
41A	(5) Secure retaining plate (1320) onto main body assembly (1475) with screws (1325). Torque screws (1325) to value specified (Ref. Table 805).(6) Carry out retaining plate/main body assembly interference check as follows (Ref. Fig. 708):(a) Check the gap between retaining plate (1320, IPL Fig.1) and main body assembly (1475) interface using a standard feeler gage.	717		

42	(b) If the gap is below the minimum dimension (0.005" (0.013 cm)), remove bypass valve assembly (1385) and retainer (1415) as specified in DISASSEMBLY. Install an additional new gasket (1420).(c) Reassemble retainer (1415) and bypass valve assembly (1385) into main body assembly (1475) and proceed with assembly as specified in steps (1) through (5), preceding.(d) Recheck the gap per instructions contained in steps(a) and (b), preceding.	719		
43	(7) Install adjustment screw (1315) into cover assembly (1330), fully in, then back out one complete turn.Assemble washer (1310) and detent cup (1305) onto adjustment screw (1315) and align detent cup line with scribe line on cover assembly (1330). Assemble washer(1300) and nut (1295), and torque nut to value specified (Ref. Table 805).	719		
44	C. Install torsion shaft assembly (995, IPL Fig.1) as follows:(1) Install new ball bearing (1005) into main body assembly (1475), using bearing assembly tool (25, Fig. 901), with sealed side of ball bearing facing outward.	719		

45	(2) Lubricate new preformed packing (1000, IPL Fig.1) with oil (1, Table 704) and assemble onto torsion shaft assembly (995, IPL Fig.1). Assemble new ball bearing (990) onto torsion shaft assembly (995) with sealed side facing outward. Assemble torsion shaft assembly (995) into main body assembly (1475).(3) Lubricate new gasket (985) with grease (3, Table 704). Install gasket (985, IPL Fig.1) and cover (975). Secure cover with screws (980) and torque to value specified (Ref. Table 805).	719		
46	D. Install metering valve assembly (1270, IPL Fig.1) as follows:(1) Install metering valve assembly (1270) into main body assembly (1475) and assemble lever assembly (940) through torsion shaft assembly (995) to locate in slot of metering valve assembly (1270).(2) Lubricate new preformed packings (1260 and 1265) with oil (1, Table 704) and assemble to metering valve sleeve assembly (1255, IPL Fig.1).	719		

47	(3) Screw metering valve sleeve assembly (1255) into retaining plate (1245) and assemble into main body assembly (1475), ensuring that metering valve assembly is aligned. Secure retaining plate (1245) with screws (1250) and torque to value specified (Ref. Table 605).(4) Assemble nut (1240, IPL Fig.1) onto metering valve sleeve assembly (1255).NOTE: Do not torque nut (1240) at this point of assembly.	720		
48	(5) Lubricate new preformed packing (1235) with oil (1, Table 704) and assemble onto adjustment screw (1230, IPL Fig.1). Lubricate adjustment screw (1230) with Oildag (4, Table 704) and assemble into metering valve sleeve assembly (1255, IPL Fig.1) (five turns only). Check that the breakaway torque of adjustment screw (1230) is within the limits specified (Ref. Table 805).	720		

49	E. Install damped bellows assembly (1200, IPL Fig.1) as follows:(1) Lubricate new gaskets (1215 and 1220) with grease (3,Table 704) on both sides and assemble onto bellows assembly (1210, IPL Fig.1).	720		
50	CAUTION:TO AVOID POSSIBLE DAMAGE TO MAIN BODY ASSEMBLY (1475) AND DAMPED BELLOWS ASSEMBLY (1200), ENSURE THAT THE FLAT ON BELLOWS ASSEMBLY (1210) IS FULLY ENGAGED IN THE SLOT OF MAIN BODY ASSEMBLY BEFORE ASSEMBLING WASHER (1060) AND NUT (1055).	720	CUIDADOS	CUIDADOS

51	(2) Assemble new bellows guide ring (1205) onto bellows assembly (1210). To ensure drainage of possible condensation, assemble sleeve (1195) with center line of holes oriented parallel to lever assembly (915). Install damped bellows assembly (1200) into main body assembly (1475). Install sleeve (1195).CAUTION: TO AVOID POSSIBLE DEFORMATION OF EVACUATED BELLOWS CONVOLUTIONS. NUT (1055) MUST BE TORQUED PRIOR TO TIGHTENING SCREWS (1115).	720		

52	(3) Secure damped bellows assembly (1200) with washer (1060) and new nut (1055). Torque nut to value specified (Ref. Table 805).(4) Assemble the manual override mechanism as follows:(a) Lubricate new preformed packings (1185 and 1190, IPL Fig.1) with oil (1, Table 704) and assemble onto body (1180, IPL Fig.1).(b) Assemble shaft (1140) into cover assembly (1120). Assemble shim(s) (1155), stop (1145) and new split pin (1150) onto shaft (1140). Do not bend the ends of split pin (1150) at this stage of assembly.	720 / 721		
53	(d) Assemble lever (1095) onto shaft (1140) with nut (1090) and pin (1085).(e) Assemble two shims (1175) onto cover assembly (1120).NOTE: Final thickness of shims (1175) will be determined at the time of preflow test.	721		
53A	CAUTION: During the following measurement process, manual override lever (1095) must be held counter clockwise against the internal stop.	721	CUIDADOS	CUIDADOS

54	(f) Assemble pin (1170A) into shaft (1140). Install drive pin (1160), then secure with spring pin (1165).(g) Assemble body (1180) onto cover assembly (1120), ensuring that locating pin (1170A) is centered.	721		
55	(h) Locally manufacture a setting sleeve as per Fig. 710. Measure Dimension 'A'. (j) Measure Dimensions 'B' and 'C' as per Fig. 710. Calculate Dimension 'X' as follows:Dimension 'X' = (Dim. Plate dimension - Dim. cm to 0.203 cm).'C' - Dim. 'B') - (Setting 'A') = 0.008" to 0.080" (0.020)(k) If limits of Dimension 'X' cannot be obtained, use alternate length pin (1170 or 1170B, IPL Fig.1).(m) Assemble set screw (1110), washer (1105) and nut (1100) onto cover assembly (1120).NOTE: Final adjustment, torquing and lockwiring will be done after calibration.	721		

56	CAUTION: IF NECESSARY TO UNSCREW AND RETORQUE NUT (10,55), ENSURE THAT SCREWS (1115) ARE LOOSENED BEFORE TORQUE IS APPLIED TO NUT (1055).(n) Assemble body (1180) and cover assembly (1120) onto main body assembly (1475) with screws (1115).NOTE: Screws (1115) will be torqued after the preflow test.	724		
57	F. Assemble and install relief valve assembly (1010, IPL Fig.1) as follows:(1) Lubricate new preformed packing (1050) with oil (1, Table 704) and assemble onto orifice (1035, IPL Fig.1).(2) Assemble ball assembly (1030) into orifice (1035) and install the orifice into main body assembly (1475).(3) Assemble nut (1045) onto plug (1015). Lubricate new preformed packing (1.040) with oil (1, Table 704) and assemble onto plug (1015, IPL Fig.1).(4) Install shim(s) (1025) and spring (1020,) into plug (1015.) and install the plug into main body assembly (1475).Torque plug (1015) and nut (1045) to values specified (Ref. Table 805).	724		

58	G. Lubricate threads of adjustment screws (955 and 970, IPL Fig.1), with grease (3, Table 704) and install, together with washers (950 and 965, IPL Fig.1) and new nuts (945 and 960), to locate in the slots of torsion shaft assembly (995).NOTE: Do not set adjustment screws (955 and 970) or torque nuts (945 and 9.60) at this point of assembly.	724		
59	H. Assemble nut (935) onto metering valve lever assembly (940).Assemble bellows lever assembly (915) through torsion shaft assembly (9951, to align with the slot in damped bellows assembly (1200). Install nut (910) and adjust fewer assemblies (915 and 940) as follows:	724		
59A	(1) Assemble operating lever setting fixture (5, Fig. 901) onto main body assembly (1475, IPL Fig. 1).(2) Adjust lever assemblies (915 and 940) to within 0.185 ± 0..010" (0.470 ± 0.025 cm) (Ref. Fig. 712).	724		

60	(3) Tighten nuts (910 and 935), using operating lever locknut wrench (26, Fig. 901) and a standard 5/64 inch Allen wrench.(4) Recheck the dimension set in step (2), preceding, after nuts (910 and 935, IPL Fig.1) have been tightened. Torque nut (910) to value specified (Ref. Table 805). Do not torque nut (935) at this stage of assembly.CAUTION: EXERCISE CARE TO PREVENT PIECES OF LOCKWIRE FROM ENTERING THE MAIN BODY ASSEMBLY DURING LOCKWIRING OPERATIONS.WARNING: WEAR SAFETY GLASSES WHEN INSTALLING LOCKWIRE.	724 / 725		
61	(5) Lock wire bellows lever assembly (915) to torsion shaft assembly (995) (Ref. Fig. 711).	725		
62	I. Adjust position of torsion shaft assembly (995, IPL Fig.1) as follows:(1) Mount flow body assembly (400) onto PowRarm (2, Fig.901), using PowRarm adapter mounting bolt (39) and PowRarm adapter (40).	725		

63	(2) Ensure that adjustment screw (1230, IPL Fig.1) and set screw (1110) are in full minimum positions, and that metering valve sleeve assembly (1255) is fully out.(3) Install metering valve travel fixture (38, Fig. 901) onto main body assembly (1475, IPL Fig.1), in adjustment screw (1285) orifice. Back off adjustment screws (955 and 970) until no further effect is noted on fixture.This ensures the adjustment screws are not loading the torsion shaft assembly (995). Set dial indicator to zero.	725		
64	(4) Using telescopic wrench (4, Fig. 901), adjust adjustment screw (955, IPL Fig.1) inward until dial indicator shows a movement of 0.028" (0.071 cm).(5) Screw adjustment screw (970) inward until a torque of 10lb. in. (1.13 N-m) is obtained. The resultant dial indicator reading must be within 0.014" to 0.018" (0.036cm to 0.045 cm).NOTE: Do not exceed 10 lb. in. (1.13 Nm) torque during this setting.	725		

65	(6) If indicated reading is out of limits, adjust by backing off both adjustment screws (955 and 970) and then resetting as per Table 701 limits. For example, if indicated reading is 0.009" (0.023 cm), adjust adjustment screw (955, IPL Fig.1) to give a movement of 0.035" (0.089 cm) on the dial indicator. This will correct the resultant indicated movement to within 0.014" to 0.018" (0.036 cm to 0.046 cm) when 10 lb. in. (1.13 N-m) torque is reapplied to adjustment screw (970).	725 /729		
66	(7) When setting is achieved, torque nuts (945 and 960) to values specified (Ref. Table 805) using special torque wrench (35, Fig. 901).WARNING: WEAR SAFETY GLASSES WHEN INSTALLING LOCKWIRE.	729		
67	(8) Lockwire and seal adjustment screws (955 and 970, IPLFig.1) using seals (40). Crimp aluminum seals, using pliers (1, Fig. 901).	729		

68	J. Lubricate new preformed packing (905, IPL Fig.1) with oil (1, Table 704) and assemble onto cover (895, IPL Fig.1).Install cover (895) and secure with screws (900). Torque screws to value specified (Ref. Table 805).	729		
69	K. Carry out metering valve assembly preflow test as follows (Ref. Fig.712:and Table 702):(1) Assemble preset adapter (2, Fig. 712) to the flow body assembly and connect an air supply as illustrated.(2) Apply air pressure as specified (Test Point l, Table 702) and zero the dial indicator.	729		
70	(3) Increase the air pressure (Test Point 2) and record the indicated metering valve travel.(4) Increase the air pressure (Test Point3) and record the indicated metering valve travel.	729		

71	(5) Reduce the air pressure (Test Point 4) and check that the metering valve travel is within the limits specified.NOTE: Adjust metering valve lever assembly (8, Fig.712) either in or out to obtain the limits specified (Test Points 2 and 3, Table 702).Repeat steps (2) through (5), preceding.	729		

72	(6) Open valve (3, Fig. 712) and vent the (Py) side of the bellows assembly to atmosphere. Maintaln the manual override lever in a counter clockwise position, apply air pressure (Test Point 5, Table 702) and check that the metering valve travel indicated on dial indicator is within specified limits. If the reading is out of limits, select shim(s) (1175, IPL Fig.1) as required to a maximum of 0.080" (0.203 cm) and record.Shim Dimension 0 0.080"(0.203 cm)	729		

73	(7) Check the clearance height of metering valve lever assembly (8, Fig. 712) as specified (Table 702).(8) Remove all special tools, fixtures and equipment.(9) Torque nut (935, IPL Fig.1) and screws (1115) to values specified (Ref. Table 805).	731		
74	L. Complete the assembly of metering valve lever assembly (940, IPL Fig.1) as follows:CAUTION : EXERCISE CARE TO PREVENT PIECES OF LOCKWIRE FROM ENTERING THE MAIN BODY ASSEMBLY DURING LOCK WIRING OPERATIONS.	731		
75	(1) Lockwire metering valve lever assembly (940) as shown in Fig. 711.(2) Lubricate new preformed packing (930, IPL Fig.1) with oil (1, Table 704) and assemble into cover (920, IPL Fig.1).(3) Assemble cover (920) onto main body assembly (1475) and secure with screws (925). Torque screws to value specified (Ref. Table 805).	731		

76	M. Lubricate new preformed packing (1290, IPL Fig.1) with oil (1, Table 704) and assemble onto adjustment screw (1285, IPL Fig.1). Install adjustment screw (1285) into main body assembly (1475). Assemble nut (1275) and washer (1280) to adjustment screw (1285). Do not torque nut (1275) at this stage of assembly.N. Install lever and shaft assembly (740) and pressurizing valve assembly (670) into pressurizing valve body (710) as follows:(1) Lubricate new preformed packings (700 and 705) with oil (1, Table 704) and assemble onto sleeve (695, IPL Fig.1).	731		
77	(2) Install sleeve (695) into pressurizing valve body (710).(3) Install valve (690), shim(s) (680) and spring (675) into sleeve (695).(4) Lubricate new preformed packing (665) with oil (1, Table704) and assemble onto retainer (650) IPL Fig.1).(5) Assemble retainer (650) onto pressurizing valve body (710) and retain with washers (660) and screws (655).Torque screws to value specified (Ref. Table 805).	731		

78	(6) Lubricate new preformed packing (760, IPL Fig.1) with oil (1, Table 704) and assemble onto shaft assembly (755, IPL Fig.1).(7) Install shaft assembly (755) through pressurizing valve body (710) and assemble lever assembly (750) onto shaftassembly (755).NOTE: To align taper pin holes correctly, ensure that lever assembly (750) arm and eccentric pin of shaft assembly (755) are diametrically opposite.	732		
79	(8) Install new split taper pin (745), using special pliers (44, Fig. 901), through lever assembly (750, IPL Fig.1) and shaft assembly (755). Ensure that both assemblies are tightly secured, then spread ends of taper pin (745).(9) Place pressurizing valve body (710) in a suitable handling container until required as per paragraph 3.P.,following.	732		

80	O. Install filter assembly (785) and elbow (765) as follows:(1) Assemble nut (770) onto elbow (765) until washer face of nut lines up with upper corner of preformed packing groove. Assemble new back-up ring (775) into nut (770).(2) Lubricate new preformed packing (780) with oil (1, Table704) and assemble into groove of elbow (765, IPL Fig.1) so that preformed packing (780) contacts nut (770) or back-up ring (775).	732		
81	(3) Assemble spring (790) onto spring guide (795) and install spring guide into main body assembly (1475).(4) Install filter assembly (785) into main body assembly (1475), with spring guide (795) leading in.	732		

82	(5) Screw elbow (765), with nut (77,0) and new preformed packing (780), into main body assembly (1475) until preformed packing (780) is in contact with main bodyassembly chamfer. Before tightening nut (770), position elbow (765) 22° off vertical towards drive shaft end (Ref. Fig. 713).(6) Tighten nut (770) and torque to value specified (Ref.Table 805).	732		
83	P. Install cut-off valve assembly (845, IPL Fig.1) as follows:(1) Install ball (855) and spring (850) into cut-off valve (865). Use spring depressor (34, Fig. 901) to depress spring (850, IPL Fig.1) and secure with spring pin (860).NOTE: Preformed packing (840) will be assembled to sleeve (835) after cut-off valve shining is completed (Ref. paragraph 4.K.).	734		

84	(2) Measure sleeve (835) counter bore depth (Ref. Fig. 717) and record in Table 703 as Dimension A.(3) Position cut-off valve assembly (845, IPL Fig.1) into sleeve (835) so that when valve (865) and sleeve (835) are installed in main body assembly (1475), open side of pin slot in cut-off valve assembly (845) will face indirection of elbow (765).(4) Lubricate new preformed packings (730 and 735) with oil (1, Table 704) and install in grooves of pressurizing valve body (710, IPL Fig.1) and main body assembly (1475).	734		
85	CAUTION: ENSURE THAT ECCENTRIC PIN OF SHAFT ASSEMBLY (755) AND SLOT OF CUT-OFF VALVE (865) ARE CORRECTLY MATED, OTHERWISE ITEMS WILL SUSTAIN PERMANENT DAMAGE (REF. FIG. 714).	734	CUIDADO	CUIDADO

85A	(5) Assemble pressurizing valve body (710) onto main body assembly (1475) and secure with washers (725) and screws (715 and 720). Torque screws to value specified (Ref.Table 805).(6) Move lever and shaft assembly (740, IPL Fig.1) and observe cut-off valve assembly (845) also moves, to ensure correct mating of items.	734		
86	Q. Assemble trim stop (480,) and throttle stop (485) onto screw (490), and assemble screw to main body assembly (1475). Ensure that throttle stop (485) locates correctly with locating pin (1495). Torque screw (490) to value specified (Ref. Table 805).	734		
86A	NOTE: Position trim stop (480, IPL Fig.1) so that it will not interfere with throttle lever movement.	734	AVISO	AVISO

87	R. Assemble bracket assembly (625) and idle reset and cut-off lever assembly (590) as follows:(1) Install bushings (610) into bracket assembly (625).(2) Assemble torsion spring (605) onto bracket assembly (625) with one end of spring hooked onto angled support of bracket assembly.(3) Assemble cut-off and idle reset lever assembly (590) onto bracket assembly (625) and position second hook of spring (605) under short arm of cut-off and idle reset lever assembly (590).	734 / 738		
88	(4) Align holes in bracket and cut-off and idle reset lever assemblies and install pin (585). Assemble washers (570 and 580) onto pin (585) and secure with new cotter pins (565 and 575). Spread ends of cotter pins.(5) Assemble pin (560) into cut-off and idle reset lever assembly (590) and secure with new cotter pin (555). Spread ends of cotter pin.	738		
89	S. Assemble bracket assembly (625) onto main body assembly (1475). Secure with washers (635) and screws (630). Torque screws to value specified (Ref. Table 805).	738		

90	T. Connect cut-off link (535, IPL Fig.1) to lever assembly (750) and cut-off and idle reset lever assembly (590) as follows: (1) Assemble spring (515) and nut (510) into clip (520).(2) Assemble cut-off link (535) through clip (520) and spring (515). Thread nut (510) onto cut-off link (535).(3) Assemble pin (505) into lever assembly (750).(4) Push clip (520) upward and hook over pin (505). Tighten nut (510) until threads of cut-off link (535) protrude through pin (505) and clip (520).	738		
91	(5) Assemble spacer (500) and nut (495) to cut-off link (535).(6) Assemble spacer (530) onto cut-off link (535) and connect to cut-off and idle reset lever assembly (590,). Install new cotter pin (525) and spread ends.	738		
	Assembly of Main Fuel Control Unit			
92	Assemble Main Fuel Control Unit and complete assembly of flow body assembly (400) as follows (Ref. IPL Fig.1):A. Remove main body assembly (1475) from PowRarm adapter (40, Fig. 901).B. Install cam (475, IPL Fig.1) into bushing (1480) in main body assembly (1475).	738 / 739		

93	C. Lubricate new preformed packings (50 and 55) with oil (1, Table 704) and assemble into grooves of flow body assembly (400, IPL Fig.1).D. Assemble drive body assembly (60) to flow body assembly (400) and secure with washers (75) and screws (65 and 70). Torque screws to value specified (Ref. Table 803).E. Reassemble Main Fuel Control Unit to PowRarm adapter (24, Fig. 901).	739		
94	F. Install throttle shaft (470, IPL Fig.1) and stop lever assembly (420) as follows:(1) Install throttle shaft (470) into main body assembly (1475).(2) Assemble vent cover (450) and secure with washers (460) and screws (455). Torque screws to value specified (Ref. Table 805).	739		
94A	WARNING: WEAR SAFETY GLASSES WHEN INSTALLING LOCKWIRE.	739	CUIDADO	CUIDADO

95	(3) Assemble stop screw (445, IPL Fig.1) and nut (440) to stop lever assembly (420) as illustrated in Fig. 715.Torque nut to value specified (Ref. Table 805) and lockwire stop screw (445, IPL Fig.1) and nut (440) to stop lever assembly (420).(4) Rotate cam (475) in a clockwise direction until a torque restraint is felt.(5) Assemble stop lever assembly (420) onto throttle shaft (470) and mesh tangs of cam (475) with slots of stop lever assembly (420).	739		
96	NOTE: Relationship of cam (475) and stop lever assembly (420) must be such that a torque restraint is felt before stop lever assembly (420) is in contact with throttle stop (485).If not, remove stop lever assembly, rotate cam180 degrees clockwise and reengage tangs of cam with stop lever assembly.	739		

97	(6) Rotate throttle shaft (470) until cotter pin holes are aligned and install new cotter pin (425). Spread ends of cotter pin.G. Assemble throttle lever link (550) into stop lever assembly (420) and install new cotter pin (405). Spread ends of cotter pin. Install other end of link through pin (560) and assemble spacer (545) and nut (540) onto throttle lever link (550).NOTE: Do not adjust nut (540) at this point of assembly. Adjustment will be made during linkage presetting.	740		
98	H. To avoid possible fouling of stop lever assembly (4, Fig. 715) against throttle stop (1), ensure minimum clearance of 0.050" (0.127 cm) between face of throttle stop (1) and highest point on un machined surface of stop lever (430, IPL Fig.1).I. Assemble stop screws (615 and 620) into bracket assembly (625) until a dimension of approximately 0.125" (0.318 cm) is obtained from underneath of screw heads to bracket assembly face. Install set screw (1440) into main body assembly (1475).	740		

99	J. Preset linkage as follows (Ref. Fig. 716):(1) Remove Main Fuel Control Unit from PowRarm adapter (24, Fig. 901).	740		
100	(2) Assemble throttle quadrant and pointer (33) and locking nut onto throttle shaft (470, IPL Fig.1) but do not tighten locking nut.(3) Assemble throttle quadrant pointer part of 33, Fig. 901) to mounting face of drive body assembly (60, IPL Fig.1).{4} Position stop lever assembly (420) against throttle stop (485).	740		
101	(5) Position throttle quadrant and pointer (33, Fig. 901) on splines of stop lever assembly with 90-degree mark as close as possible to pointer.(6) Tighten locking nut of throttle quadrant.NOTE: Adjust degree plate of throttle quadrant to align90-degree mark with pointer, if necessary.	740		
102	(7) Move stop lever assembly (420, IPL Fig.1} to abut set screw (1440) (Ref. A, Fig. 71.6) and adjust set screw, if necessary, to align minus five-degree mark with pointer.	740		

103	(8) Position cable arm (the arm with two holes in the end) of cut-off and idle reset lever assembly (590, IPL Fig.1) at 25° throttle angle (Ref. B, Fig. 716) to point toward the throttle shaft (470, IPL Fig. 1) and perpendicular to bracket assembly (625) mounting face.(9) Release the cut-off and idle reset lever assembly (590) and set stop lever assembly (420) at eight-degree position. Lock in position using lock screw of throttle quadrant (33, Fig. 901) pointer.	743		
104	(10) Adjust nut (510, IPL Fig.1) (Ref. D, Fig. 716) until lever assembly (750, IPL Fig.1) is parallel to bracket assembly (625) mounting face.(11) Set stop lever assembly (420) to 16 degree position and lock in position.	743		
105	K. Shim sleeve (835) and cut-off valve assembly (845) as follows (Ref. Fig. 717):(1) Ensure that sleeve (3) is correctly bottomed in main body assembly (6) bore.	743		

106	(2) Ensure that Dimension A is entered be side corresponding sleeve dimension in Table 703 [Ref. paragraph 3.P.(2)].(3) Measure differential height (Dimension B) from top of cut-off valve assembly (5, Fig. 717) and top of sleeve (3).NOTE: If top of cut-off valve assembly (5) is above top of sleeve (3), Dimension B becomes a plus dimension and if below, it becomes a minus dimension.	743		
107	(4) Enter Dimension B in adjacent column to Dimension A in Table 703.(5) Observe clearance height (Dimension C) that top of sleeve (3, Fig. 717) must be above top of cut-off valve assembly (5).(6) Calculate shim thickness (Dimension D) required by either adding (plus) or subtracting (minus) Dimension B to/fromDimension C.(7) Remove sleeve (3), lubricate new preformed packing (7) with oil (1, Table 704) and assemble onto sleeve (3, Fig.717).	743		

108	(8) Install required thickness of shims (4) into main body assembly (6) bore and reinstall sleeve (3), ensuring that sleeve is correctly bottomed in main body assembly (6) bore.(9) Move throttle quadrant (33, Fig. 901) to between 8 and 10-degree position and lock in position.(10) Measure difference in height from top of sleeve (3, Fig.717) to top of cut-off valve assembly (5) (cut-off valve higher than sleeve).	745		
109	(11) Add thickness of shim(s) (2), equal to differential height, to top of sleeve (3).(12) Assemble seat (1) on top of shim(s) (2).NOTE: Seat (1) must be presoaked in calibration fluid (Specification MIL-C-7024, Type II) for a period of eight hours prior to assembly.	745		
110	(13) Lubricate new preformed packings (815 and 820, IPL Fig.1) with oil (1, Table 704) and assemble onto elbow adapter (800, IPL Fig.1).(14) Assemble plate (810) and screws (805) to retain elbow adapter (800). Torque screws (805) to value specified (Ref. Table 805).	745		

111	L. Complete linkage presetting as follows:(1) Move throttle quadrant and pointer (33, Fig. 901) to minus five-degree position.(2) Adjust nut (495, IPL Fig.1) (Ref. E, Fig. 716) to give a clearance of 0.005" to 0.0010" (0.013 cm to 0.0025 cm) between nut (495, IPL Fig.1) and spacer (500).NOTE: Spacer (500) must be in contact with clip (520) when measuring clearance.	745		
112	(3) Adjust stop screw (620) (Ref. F, Fig. 716) to just contact cut-off and idle reset lever assembly (590).(4) Remove throttle quadrant and pointer (33, Fig. 901).(5) Assemble spacer (25, IPL Fig.1) and nut (415) onto stop lever assembly (420). Tighten hand tight only until cotter pin holes align and install new cotter pin (410). Spread ends of cotter pin sufficient to prevent cotter pin from falling out.	745		

113	M. Lubricate new preformed packing (890,) with oil (1, Ref. Table 704) and assemble onto plug (885, IPL Fig.1). Screw plug into main body assembly (1465) and torque to value specified (Ref. Table 805).N. Assemble washer (880, IPL Fig.1) and screw (875) onto top of main body assembly (1475) and torque screw to value specified (Ref. Table 805).	746		
114	O. Assemble identification plate (30, IPL Fig.1) to Main Fuel Control Unit and secure with screws (35). Torque screws to value specified (Ref. Table 803).P. Stamp and attach new rubber assembly date decal (45, IPL Fig.1) as follows:(1) Stamp assembly date on rubber assembly date decal (45) using Meyer cord fast drying black imprint ink (Ref. Table 401).	746		
115	(2) Clean decal mounting area of Main Fuel Control Unit thoroughly with cleaning solvent FED. SPEC. P-D-680 (Ref.Table 401) to remove any oil film.(3) Apply a thin uniform film of Meyercord Cement No. 444-HX (Ref. Table 401) to	746		

115A	decal mounting area of Main Fuel Control Unit.(4) Submerge rubber assembly date decal (45, IPL Fig.1) in water for approximately 30 seconds, then slide decal from paper backing, face up, onto cemented surface.	746		
116	(5) Press and roll out all air bubbles and excess cement from under decal. Remove excess cement from housing with a cloth moistened with cleaning solvent FED. SPEC. P-D-680 (Ref. Table 401).(6) Dry at room temperature for 15 minutes, then apply final coating of Meyercord Cement No. 444-BX over decal.Ensure final coating of cement extends 1/16" to 1/8"(0.1588 cm to 0.3175 cm) beyond all edges of decal.	746		
117	Q. Install all shipping caps, covers and plugs (5, 10, 15 and20, IPL Fig. 1).NOTE: Do not lockwire and seal unit until completion of testing. For location of lockwire and seals (40),refer to TESTING AND TROUBLESHOOTING, paragraph 4.	746		

Obs: